

TRANSPORTATION TECHNOLOGY

Transportation affects our lives in a multitude of ways. Transportation systems move raw materials to manufacturers and finished products to consumers locally, nationally, and globally. Individuals use transportation systems every day for business, work, and pleasure.

Transportation technology courses provide students with opportunities to understand transportation systems from the perspective of either the consumer or the service provider. The range of courses enables students to study both vehicle ownership and vehicle maintenance, and to develop skills and prepare for careers in the servicing and repair of vehicles, aircraft, and/or watercraft. In addition, students will develop the Essential Skills and work habits that are important for success in the transportation industry.

The list of approved emphasis areas for transportation technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Transportation Technology, Grade 11

College Preparation

TTJ3C

This course enables students to develop technical knowledge and skills as they study, test, service, and repair engine, electrical, suspension, brake, and steering systems on vehicles, aircraft, and/or watercraft. Students will develop communication and teamwork skills through practical tasks, using a variety of tools and equipment. Students will develop an awareness of environmental and societal issues related to transportation, and will learn about apprenticeship and college programs leading to careers in the transportation industry.

Prerequisite: None

A. TRANSPORTATION TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of the fundamental principles of engines and their service, repair, and maintenance;
- A2.** demonstrate an understanding of basic electrical and electronic circuits and their components;
- A3.** identify the function and explain the operation of the major systems and components of vehicles, aircraft, and/or watercraft;
- A4.** demonstrate accurate and appropriate use of technical and mathematical knowledge and skills in the study of transportation technology.

SPECIFIC EXPECTATIONS

A1. Understanding Engines

By the end of this course, students will:

- A1.1** explain terminology describing internal combustion engines (*e.g., top dead centre, over-head camshaft*), cylinder configurations (*e.g., V6, in-line*), and types of measurement (*e.g., bore, stroke, cylinder displacement*) in terms of engine operation;
- A1.2** describe the design and construction of various components of an internal combustion engine (*e.g., camshafts, pistons, crankshafts, rotors, valves, turbines*);
- A1.3** describe the service procedures required to rebuild, repair, and maintain engines (*e.g., use of correct torque procedures, use of plastigauge to check bearing clearance, use of feeler gauges and micrometers to check clearances, use of chemical compounds [sealants, thread-locking compounds]*);
- A1.4** explain the principles on which the operation of fuel, lubrication, and coolant systems is based (*e.g., fuel: volatility; lubrication: viscosity; coolant: heat transfer*);
- A1.5** explain how engine timing (*e.g., ignition timing, valve timing, mechanical fuel injection timing*) is used to achieve optimal engine performance;
- A1.6** describe the parts, operation, and care of a variety of precision measuring tools (*e.g., micrometer, vernier caliper, torque wrench, dial indicator, bore gauge*) and demonstrate proper use of these tools;

- A1.7** describe common procedures for inspecting engine components for wear and faults (*e.g., check cylinder for taper, check cylinder head for warpage*) and demonstrate accurate use of these procedures.

A2. Understanding Electrical and Electronic Circuits and Components

By the end of this course, students will:

- A2.1** describe the fundamental concepts and laws related to the flow of electricity (*e.g., flow of electrons, magnetic fields, Ohm's law, Kirchhoff's laws*) that underlie the electrical components and systems found in vehicles, craft, and powered equipment;
- A2.2** define electrical units of measure (*e.g., volts, amps, ohms*) and demonstrate the ability to read wiring schematics (*e.g., wire size and colour, symbols*);
- A2.3** define the fundamentals of electronic circuits and components (*e.g., on-board computers, diodes, transistors, light-emitting diodes [LEDs]*);
- A2.4** explain the consequences of open, short, ground, and unintentional ground circuits (*e.g., electrical surges, voltage drop, voltage spike*);
- A2.5** describe basic procedures involved in circuit repair (*e.g., use of heat shrink, use of solder and solderless connections*);

A2.6 describe the operation of various types of protection devices (*e.g., fuses, relays, circuit breakers, fusible links*);

A2.7 identify various types of batteries (*e.g., gel type, lead acid, lithium ion*) and describe their construction and applications;

A2.8 describe the design and applications of various types of electrical systems (*e.g., ignition systems, lighting systems, starting systems, charging systems*).

A3. Understanding Major Systems and Components

By the end of this course, students will:

A3.1 identify the function and explain the operation of the major components of various types of steering/control systems;

A3.2 identify the function and explain the operation of the major components of various types of suspension systems (*e.g., coil springs, struts, shocks, air springs*);

A3.3 identify the function and explain the operation of the major components of various types of brake systems (*e.g., mechanical: emergency brake cable; hydraulic: master cylinder; pneumatic: air brake chamber*);

A3.4 identify various body components of vehicles, aircraft, and/or watercraft (*e.g., fenders, doors, wheel wells, rocker panels, hull, fuselage, roll-over protection system*);

A3.5 identify and describe common types of body and frame construction (*e.g., unibody construction, subframe and structural assemblies, hull construction, airframe construction*) in vehicles or craft.

A4. Technological and Mathematical Literacy

By the end of this course, students will:

A4.1 demonstrate appropriate use of technical terminology related to vehicle or craft systems and components and to processes, tools, and equipment commonly used in the transportation industry;

A4.2 define and accurately calculate various measurements related to a vehicle or craft (*e.g., engine displacement, voltage drop*);

A4.3 prepare reports (*e.g., work orders, journals, parts lists*) for a variety of audiences, using appropriate technical language and relevant technical and mathematical knowledge and skills.

B. TRANSPORTATION TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate an understanding of engine operation and repair by performing a variety of service operations;
- B2.** demonstrate the ability to test and repair basic electrical circuits safely and correctly;
- B3.** demonstrate the ability to service and repair steering/control, suspension, brake, and body systems;
- B4.** develop appropriate solutions to transportation technology challenges and/or repair problems.

SPECIFIC EXPECTATIONS

B1. Engine Service and Repair

By the end of this course, students will:

- B1.1** access engine specifications, trouble charts, and/or diagnostic procedures from reliable information sources (*e.g., shop manuals, online information, manufacturers' information*) and use them as required to service or repair engines;
- B1.2** use a variety of hand, power, and specialty tools safely and correctly to perform basic maintenance and repair procedures (*e.g., maintenance: oil change, tune up; repair: replace the timing belt, repair a fluid leak*) on several types and styles of engines, to manufacturers' specifications;
- B1.3** use a variety of tools and equipment (*e.g., fuel pressure/vacuum gauge, compression tester, coolant pressure tester, stethoscope, manometer*) safely and correctly to diagnose basic engine condition;
- B1.4** demonstrate correct diagnostic and repair procedures in disassembling and reassembling an engine (*e.g., perform a visual inspection, measure component wear, replace or refurbish components, use proper assembly torque and sequence operations*).

B2. Electrical Circuit Testing and Repair

By the end of this course, students will:

- B2.1** demonstrate a working knowledge of inspection and testing of circuits (*e.g., use test lights or*

multimeters when required, follow correct procedures in performing a voltage drop test);

- B2.2** use a variety of test equipment (*e.g., starter circuit tester, charging system analyser*) to determine continuity and measure voltage, amperage, and resistance in various electrical circuits (*e.g., starting circuits, charging circuits, lighting circuits*);
- B2.3** perform repairs on electrical circuits (*e.g., terminal repair, wiring repair*) safely and correctly.

B3. Service and Repair of Steering/Control, Suspension, Brake, and Body Systems

By the end of this course, students will:

- B3.1** perform an inspection of steering/control, suspension, brake, and/or body components, and report on their condition, referring to appropriate information sources (*e.g., manufacturers' specifications*) and using appropriate technical language and/or illustrations;
- B3.2** perform routine and/or scheduled service procedures on steering/control, suspension, and brake systems (*e.g., lubrication, brake service, suspension inspection, belt and track adjustment, tire service, fluid change*) safely and correctly, using appropriate service information;
- B3.3** perform service and repair procedures on various body components, using appropriate fasteners and bonding agents (*e.g., bolts, welds, rivets, clips, adhesives*).

B4. Challenges and Repair Problems

By the end of this course, students will:

- B4.1** demonstrate the ability to use a problem-solving process to address a given transportation technology challenge (see pp. 21–23);
- B4.2** use relevant mathematical skills and apply appropriate scientific concepts to understand a challenge or perform repairs (*e.g., math skill: calculate clearances; concept applied: hydraulics; challenge: vehicle height modification*);
- B4.3** systematically troubleshoot basic repair problems on a vehicle or craft by using an appropriate diagnostic procedure (*e.g., gather information, perform tests, generate solutions, apply an appropriate solution, test the results*);
- B4.4** identify issues related to a challenge or repair task (*e.g., cost, availability of parts or materials, time required*) and explain how these issues could affect the response to the challenge or the performance of the repair;
- B4.5** demonstrate the safe and correct use of a variety of soldering, heating, cutting, and welding techniques when performing tasks related to a challenge or repair.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of environmental issues related to the use of materials and procedures in the service, repair, and recycling of vehicles or craft;
- C2.** demonstrate an understanding of the relationship between society, vehicle ownership, and various aspects of transportation technology.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** describe the legislative requirements concerning the use of environmentally friendly products in the repair and service of vehicles or craft, and explain the costs and benefits of using such products;
- C1.2** explain the importance of the proper processing of waste products (*e.g., batteries, used oil, antifreeze, refrigerant, tires*) as a means of minimizing the environmental impact of the transportation industry;
- C1.3** describe appropriate actions to be taken in the event of a spill of waste products (*e.g., gasoline, antifreeze*), and demonstrate the ability to safely implement such actions (*e.g., outline the steps described in an emergency action plan and carry them out*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** explain the pros and cons of personal vehicle ownership versus public transit in terms of their effects on society;
- C2.2** describe the effects that increasing transportation costs (*e.g., rising fuel costs, highway tolls, environmental taxes*) have on society;
- C2.3** explain how current societal needs and preferences influence transportation technology (*e.g., by creating a greater demand for vehicles with improved safety features, fuel efficiency, and emission levels*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate the use of professional work practices and procedures and compliance with occupational health and safety regulations and standards;
- D2.** describe career opportunities in the transportation industry and the education and training required for them.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify and explain the importance of legislation and regulations related to procedures and operations used in transportation technology facilities (e.g., *Occupational Health and Safety Act [OHSA]; regulations and standards outlined in the Workplace Hazardous Materials Information System [WHMIS]; Apprenticeship and Certification Act [ACA]*);
- D1.2** demonstrate good housekeeping and safety practices in the work environment (e.g., *cleaning up spills and leaks, keeping areas clean and clear of obstructions*);
- D1.3** use protective clothing and equipment (e.g., *eye protection, gloves, breathing mask*) as required to ensure their own and others' safety in the work environment;
- D1.4** identify potential health risks (e.g., *brake dust, fumes from brake fluid and brake cleaner*) when servicing vehicles or craft, and demonstrate the use of safe procedures to mitigate these hazards (e.g., *use appropriate ventilation and breathing protection*);
- D1.5** describe and demonstrate the ability to follow appropriate safety precautions required for new technologies when working on vehicles, craft, or power equipment (e.g., *precautions regarding high current and voltage, capacitor discharge rate of supplemental restraint systems, extreme temperature of exhaust systems*);
- D1.6** demonstrate an understanding of professional responsibilities in the transportation industry with regard to personal and public safety (e.g., *quality workmanship, integrity, customer service, compliance with manufacturers' standards*).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** describe a variety of career opportunities in the transportation industry (e.g., *apprenticeship/trade, parts person, service manager/writer*) and the education and training required for them;
- D2.2** describe the regulations regarding restricted skill sets in the Apprenticeship and Certification Act (available at www.e-laws.gov.on.ca);
- D2.3** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the transportation industry (e.g., *mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- D2.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the transportation industry, as identified in the Ontario Skills Passport (e.g., *job task planning and organizing, decision making, finding information*);
- D2.5** demonstrate an understanding of and apply the work habits that are important for success in the transportation industry, as identified in the Ontario Skills Passport (e.g., *working safely, reliability, initiative*);
- D2.6** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in transportation technology (e.g., *Passport to Safety certificate, skills checklist, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.